# Xiaoyu Ou

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**Available for a postdoctoral position from June 2027.**

## Education

**University of Bristol, Bristol, UK** 2023–Present
PhD in Electrical and Electronic Engineering
Fully funded by the University of Bristol
Thesis: *Transceiver Design and Resource Allocation for Finite-Blocklength Multiuser MIMO Communications.*
Supervisors: Dr. Shuping Dang and Prof. Angela Doufexi.
Expected thesis submission: November 2026; expected viva: before April 2027.

**Chongqing University of Posts and Telecommunications, Chongqing, China** 2020–2023
MSc in Information and Communication Engineering
Supervisor: Prof. Xianzhong Xie.

**Chongqing University of Posts and Telecommunications, Chongqing, China** 2016–2020
BEng in Electronic Science and Technology

## Research Experience

**Finite-Blocklength Communications and URLLC**

- Characterizing achievable rates and decoding error probabilities in short-packet wireless communications.
- Investigating reliability, latency, and throughput trade-offs and their implications for transmission design in ultra-reliable low-latency communications systems.
- Optimizing coding rates, blocklength, and power allocation to improve reliability and transmission efficiency.

**Multiuser MIMO and Interference Management**

- Designing precoding and receive-combining schemes for multiuser MIMO systems to separate user signals and reduce inter-user interference.
- Investigating the effects of spatial correlation, polarization, and imperfect channel state information on interference management and transceiver design.
- Developing low-complexity resource allocation algorithms for multiuser MIMO systems with rate-splitting multiple access and non-orthogonal multiple access.

**SDR Prototyping and Over-the-Air Validation**

- Implementing multiuser algorithms on USRP X310 SDR testbeds using Python/UHD and MATLAB.
- Developing device synchronization, channel estimation, and multiuser precoding.
- Evaluating performance through over-the-air experiments under practical wireless conditions.

## Teaching and Academic Service

**Teaching Assistant, University of Bristol** Feb. – May 2026
Helped supervise the undergraduate group project
*Covert Communications via Motion States of Unmanned Aerial Vehicle.*

**Journal reviewer:** IEEE Trans. Commun.; IEEE Trans. Wireless Commun.; IEEE Trans. Veh. Technol.; IEEE Commun. Lett.; IEEE Wireless Commun. Lett.

## Honors and Awards

Outstanding Master's Thesis of Chongqing 2024
*Chongqing Municipal Education Commission*

University of Bristol PhD Scholarship in Electrical and Electronic Engineering 2023
*University of Bristol*

Outstanding Graduate of Chongqing 2023
*Chongqing Municipal Education Commission*

National Scholarship 2022
*Ministry of Education of the People's Republic of China*

## Technical Skills

**Programming:** MATLAB and Python for numerical simulation; Python for algorithm implementation and SDR testbed control.

**Wireless signal processing:** Multiuser MIMO transceiver design; RSMA/NOMA interference management; channel estimation; finite-blocklength performance analysis.

**Mathematical modeling and optimization:** Communication-system modeling and optimization problem formulation; convex optimization; low-complexity resource-allocation algorithms.

**SDR experimentation:** USRP X310 testbed development using Python/UHD; device synchronization; over-the-air testing.

## Publications

### First-Author Journal Articles

**[J1]** **X. Ou**, S. Dang, and A. Doufexi, "Uplink RSMA with Dual-Polarized Multi-User MIMO over Correlated Rician Channels in Finite Blocklength Regime," *IEEE Transactions on Communications*, vol. 74, pp. 14516–14531, 2026.

**[J2]** **X. Ou**, S. Dang, and A. Doufexi, "Reliable Uplink Transmission for Low-Throughput Massive MIMO Networks With Finite Blocklength," *IEEE Transactions on Communications*, vol. 74, pp. 2716–2730, 2026.

**[J3]** **X. Ou**, S. Dang, Z. Ren, and A. Doufexi, "Uplink Power Control for Massive MIMO-NOMA With Group-Level SIC in Massive URLLC Services," *IEEE Transactions on Wireless Communications*, vol. 24, no. 10, pp. 8505–8519, Oct. 2025.

**[J4]** **X. Ou**, S. Dang, Z. Ren, and A. Doufexi, "Joint User Scheduling, Power Allocation, and Rate Control for MC-RSMA in URLLC Services," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 6, pp. 9210–9223, Jun. 2025.

**[J5]** **X. Ou**, S. Dang, and A. Doufexi, "Sum Spectral Efficiency Maximization for Uplink Multi-Layer RSMA With Finite SIC," *IEEE Wireless Communications Letters*, vol. 13, no. 12, pp. 3409–3413, Dec. 2024.

**[J6]** **X. Ou**, X. Xie, H. Lu, and H. Yang, "Channel Blocklength Minimization in MU-MISO Nonorthogonal Multiple Access for URLLC Services," *IEEE Systems Journal*, vol. 18, no. 1, pp. 36–39, Mar. 2024.

**[J7]** **X. Ou**, X. Xie, H. Lu, and H. Yang, "Resource Allocation in MU-MISO Rate-Splitting Multiple Access With SIC Errors for URLLC Services," *IEEE Transactions on Communications*, vol. 71, no. 1, pp. 229–243, Jan. 2023.

**[J8]** **X. Ou**, X. Xie, H. Lu, H. Yang, and H. Tang, "Energy-Efficient Resource Allocation for Short Packet Transmission in MISO Multicarrier NOMA," *IEEE Transactions on Vehicular Technology*, vol. 71, no. 12, pp. 12797–12810, Dec. 2022.

### Collaborative Journal Articles

**[J9]** W. B. Abbas, **X. Ou**, G. Hilton, S. Dang, A. Doufexi, and M. A. Beach, "Antenna Selection and Array Directivity Tradeoff in Massive MIMO Systems With Uniform Rectangular Arrays," *IEEE Transactions on Communications*, vol. 74, pp. 7285–7299, 2026.

**[J10]** W. Wang, **X. Ou**, Z. Ren, W. B. Abbas, S. Dang, A. Doufexi, and M. A. Beach, "Analysis on Energy Efficiency of RIS-Assisted Multiuser Downlink Near-Field Communications," *IEEE Transactions on Communications*, vol. 73, no. 11, pp. 12441–12455, Nov. 2025.

**[J11]** Y. Chen, H. Yang, **X. Ou**, Y. Jiang, and Z. Xiong, "Anti-Jamming Resource Allocation for Integrated Sensing and Communications Based on Game-Guided Reinforcement Learning," *IEEE Wireless Communications Letters*, vol. 14, no. 1, pp. 223–227, Jan. 2025.

**[J12]** X. Xie, **X. Ou**, H. Lu, and Q. Huang, "Joint Uplink and Downlink Resource Allocation in NOMA for End-to-End URLLC Services," *IEEE Communications Letters*, vol. 25, no. 12, pp. 3942–3946, Dec. 2021.

### Conference Paper

**[C1]** **X. Ou**, S. Dang, and A. Doufexi, "Dual-Polarized MIMO for Uplink Rate-Splitting Transmission with Finite Blocklength," in *Proc. IEEE 36th International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, Istanbul, Türkiye, Sep. 2025, pp. 1–6. (Invited paper.)

### Manuscript Under Review

**[S1]** **X. Ou**, S. Dang, and A. Doufexi, "On Mismatch of Sum-Rate Maximization Between Infinite and Finite Blocklength Regimes," under review at *IEEE Transactions on Signal Processing*, 2026.